Form A

2. (a) H_0: age group and weekly church attendance are independent
H_a: age group and weekly church attendance are associated

(b) The smallest expected frequency is $\frac{(111)(105)}{455} = 25.62$, well beyond 5.
The requirements are met.

(c)

	Attend weekly	Not weekly
Observed	24	87
Expected	25.62	85.38

$\chi^2 = \frac{(24-25.62)^2}{25.62} + \frac{(87-85.38)^2}{85.38}$ + 4 other terms

30-49 contribution = $\underline{0.1332}$

(d) 1 - pchisq(3.6263, df = 2)

(e) We fail to reject that age group and weekly church attendance are independent

3. (a) gf_point(rent ~ value, data = rentalProperties)

(b) The residuals-vs-fitted plot (center graph) looks reasonably scattered about the zero line without a pattern, and the normal quantile plot is fairly straight (except at one end). The SLM seems (mostly) confirmed.

(c) The point estimate is $b_1 = 0.02474$ and $SE_{b_1} = 0.00422$
The 88% CI is

$$b_1 \pm t_c(SE_{b_1}) = 0.02474 \pm (1.602)(0.00422), \text{ or } (0.0180, 0.0315)$$

(d) summary(lm(rent ~ value, data = rentalProperties))

(e) $\widehat{rent} = 5.2289 + 0.02474(value)$ (with both rent, value measured in the thousands)

(f) $\widehat{rent} = 5.2289 + 0.02474(220) = 10.672$,
or \$10,672.00.

(g) cor(rent ~ value, data = rentalProperties)
percent of variation explained = $R^2 = (0.7368)^2$, or $\underline{54.29\%}$.

4. (a) A normal plot is a device to help determine if a quantitative variable seems to have a normal distribution.
It is unnecessary, here, as we were told in the problem description that the populations are normal.

(b) $\frac{S_{max}}{S_{min}} = \frac{1.460032}{1.04985} = 1.391 < 2$ The rule of thumb is <u>satisfied</u>.

(c) $df1 = 4 - 1 = 3$

$df2 = (22+26+23+29) - 4 = 96$

$MSG = \frac{16.8949}{3} = 5.632$

$\frac{5.632}{MSE} = 3.5547 \Rightarrow MSE = 1.584$

$\frac{SSE}{96} = 1.584 \Rightarrow SSE = 152.1$

(d) 1 - pf(3.5547, df1 = 3, df2 = 96)

(e) Tukey pairwise comparisons are conducted only when the null hypothesis of ANOVA has already been rejected. Each line above is specifically about the difference of two group means — the first line is about $\mu_B - \mu_A$, the second about $\mu_C - \mu_A$, and so on. The one significant p-value at $\alpha = 0.05$ is line 1, allowing us to reject $H_0: \mu_B - \mu_A = 0$ in favor of $H_a: \mu_B - \mu_A \neq 0$.

Conclusions at the 5% level:

$\mu_A \neq \mu_B$ (the only one)

(f) TukeyHSD(len ~ grp, data = lengthByGrp)